



Tennessee Valley Authority, Post Office Box 2000, Spring City, Tennessee 37381

May 24, 2018

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Watts Bar Nuclear Plant, Units 1 and 2
Facility Operating License Nos. NPF-90 and NPF-96
NRC Docket Nos. 50-390 and 50-391

Subject: **Licensee Event Report 390/2018-003-00, Control Room Emergency
Ventilation System Inoperable due to Main Control Room Door Being Left
Open**

This submittal provides Licensee Event Report (LER) 390/2018-003-00. This LER provides details concerning a condition where the Main Control Room envelope door was left open by plant personnel. This condition is being reported as an event that could have prevented fulfillment of a safety function in accordance with 10 CFR 50.73(a)(2)(v).

There are no regulatory commitments contained in this letter. Please direct any questions concerning this matter to Kim Hulvey, WBN Licensing Manager, at (423) 365-7720.

Respectfully,

A handwritten signature in black ink, appearing to read "Paul Simmons", with a long horizontal line extending to the right.

Paul Simmons
Site Vice President
Watts Bar Nuclear Plant

Enclosure
cc: See Page 2

U.S. Nuclear Regulatory Commission
Page 2
May 24, 2018

cc (Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Watts Bar Nuclear Plant



LICENSEE EVENT REPORT (LER)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NE08-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Watts Bar Nuclear Plant, Unit 1	2. Docket Number 05000390	3. Page 1 OF 5
--	-------------------------------------	--------------------------

4. Title Control Room Emergency Ventilation System Inoperable due to Main Control Room Door Being Left Open

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	26	2018	2018	- 003	- 00	05	24	2018	Watts Bar Nuclear Plant, Unit 2	05000391
									Facility Name	Docket Number

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. Licensee Contact for this LER	
Licensee Contact Dean Baker, Licensing Engineer	Telephone Number (Include Area Code) 423-452-4589

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	VI	DR	NONE	Y					

14. Supplemental Report Expected		15. Expected Submission Date	
☐ Yes (If yes, complete 15. Expected Submission Date)	☒ No	Month	Day

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

At 1058 Eastern Daylight Time (EDT) on March 26, 2018, a Main Control Room (MCR) alarm was received for low control room positive pressure. At 1100 EDT, a Control Room Envelope (CRE) door was found ajar and immediately closed. Technical Specification Limiting Condition for Operation (LCO) 3.7.10, Control Room Emergency Ventilation System (CREVS), was declared not met for both trains and Condition B entered. At 1100 EDT on March 26, 2018, the alarm cleared, CREVS was declared operable and LCO 3.7.10, Condition B was exited.

Later that day at 1839 EDT, a MCR alarm was received for low control room positive pressure. At 1840 EDT, a CRE door was found ajar and immediately closed. Technical Specification LCO 3.7.10 for CREVS, was declared not met for both trains and Condition B entered. At 1840 EDT on March 26, 2018, the alarm cleared, CREVS was declared operable and LCO 3.7.10, Condition B was exited.

These events were caused by human performance errors when individuals traversing the control building complex failed to confirm the MCR envelope boundary door was properly latched. A worn latching component of the MCR envelope door contributed to this event, and was replaced. A site stand-down was conducted to brief and reaffirm to site personnel on the requirements for door closure throughout the plant. An engineering change is in development to magnetically lock the MCR door and to provide indication when the door is not fully shut and latched.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Watts Bar Nuclear Plant, Unit 1	05000390	2018	- 003	- 00

NARRATIVE**I. Plant Operating Conditions Before the Event**

Watts Bar Nuclear Plant (WBN) Unit 1 and Unit 2 were at 100 percent rated thermal power (RTP) .

II. Description of Event**A. Event Summary**

At 1058 Eastern Daylight Time (EDT) on March 26, 2018, a Main Control Room (MCR) alarm was received for low control room positive pressure. At 1100 EDT, a Control Room Envelope (CRE) door {EIIIS:DR} was found ajar and immediately closed. Technical Specification Limiting Condition for Operation (LCO) 3.7.10, Control Room Emergency Ventilation System (CREVS){EIIIS:VI}, was declared not met for both trains and Condition B entered. At 1100 EDT on March 26, 2018, the alarm cleared, CREVS was declared operable and LCO 3.7.10, Condition B, was exited.

Later that day at 1839 EDT, a MCR alarm was received for low control room positive pressure. At 1840 EDT, a CRE door was found ajar and immediately closed. Technical Specification LCO 3.7.10 for CREVS, was declared not met for both trains and Condition B entered. At 1840 EDT on March 26, 2018, the alarm cleared, CREVS was declared operable and LCO 3.7.10, Condition B was exited.

This event is being reported to the Nuclear Regulatory Commission (NRC) under 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented fulfillment of a safety function needed to mitigate the consequences of an accident.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

A worn latching component contributed to this condition.

C. Dates and approximate times of occurrences

<u>Date</u>	<u>Time (EDT)</u>	<u>Event</u>
3/26/18	1058	MCR alarm was received for low control room positive pressure. TS 3.7.10 was declared not met for both trains and Condition B entered
3/26/18	1100	CRE door was found ajar and immediately closed. TS 3.7.10 Condition B exited with cleared alarm
3/26/18	1839	MCR alarm was received for low control room positive pressure. TS 3.7.10 was declared not met for both trains and Condition B entered
3/26/18	1840	CRE door was found ajar and immediately closed. TS 3.7.10 Condition B exited with cleared alarm.

D. Manufacturer and model number of each component that failed during the event

The door involved in this event was fabricated by TVA.

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Watts Bar Nuclear Plant, Unit 1	05000390	2018	- 003	- 00

NARRATIVE**F. Method of discovery of each component or system failure or procedural error**

Plant alarms indicated a loss of MCR positive pressure. The response procedure for low MCR pressure requires that the MCR doors be checked for proper closure, at which time door C054 was found open.

G. Failure mode, mechanism, and effect of each failed component

A worn component for door C054 made the door difficult to close.

H. Operator actions

Upon receipt of the alarms, operations personnel promptly closed the MCR boundary door.

I. Automatically and manually initiated safety system responses

The MCR low pressure alarm properly actuated when the MCR door was left open.

III. Cause of the Event**A. Cause of each component or system failure or personnel error**

A worn component for door C054 was replaced as a result of these events, but this equipment deficiency would not have prevented door closure if properly closed by the individuals operating the MCR door.

B. Cause(s) and circumstances for each human performance related root cause

Both events were the result of individuals operating the boundary door failing to properly close the door and confirm its closure (lack of attention to task).

IV. Analysis of the Event

The CRE is required to be operable in Modes 1 through 6. Operability requires integrity of the CRE such that it will have a low unfiltered in-leakage during accident conditions to maintain the dose to operators within the requirements of 10 CFR 50, General Design Criterion 19. The TS's allow the CRE boundary to be opened intermittently under administrative control, normally to allow routine personnel ingress and egress from the control room envelope. Administrative controls in the case of boundary doors are that an individual is in control of the door when it is opened.

On March 26, 2018, on two occasions individuals traversing the control building complex left the Unit 2 MCR boundary door C054 ajar. This resulted in operations personnel entering TS LCO 3.7.10, CREVS, for one or more CREVS trains inoperable due to an inoperable CRE boundary. Low positive pressure (less than 0.125 inches of water gauge WG) in the control room for 90 seconds results in a control room alarm. Upon receipt of the alarm, operations personnel promptly closed the CRE door. For these events, the CRE boundary was open approximately three minutes. Engineering evaluation of these events conclude that General Design Criteria (GDC) 19 dose limits to operators would not be exceeded when considering closure of the MCR door for accidents analyzed in the Updated Final Safety Analysis Report.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Watts Bar Nuclear Plant, Unit 1	05000390	2018	- 003	- 00

NARRATIVE**V. Assessment of Safety Consequences**

A review of this event indicates, when considering the actual system capability and the response of equipment and personnel, a loss of safety function capable of impacting public health and safety did not occur with respect to the Control Room. This equipment is not analyzed in the site specific probabilistic risk assessment (PRA), but the impact of this door on an accident would be very small.

- A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

The balance of the CRE equipment designed to protect the pressure boundary remained operable.

- B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

Not applicable.

- C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from the discovery of the failure until the train was returned to service

In both cases, the MCR envelope door was closed within about two minutes of receipt of the MCR alarm.

VI. Corrective Actions

These events were entered into the Tennessee Valley Authority (TVA) Corrective Action Program and are being tracked under Condition Reports (CRs) 1399593 and 1399747.

- A. Immediate Corrective Actions

Operations personnel promptly closed the MCR door. A worn latching component of the MCR door was replaced.

- B. Corrective Actions to Prevent Recurrence or to reduce probability of similar events occurring in the future

A site stand-down was conducted to brief and reaffirm to site personnel on the requirements for door closure throughout the plant. An engineering change is in development to magnetically lock the MCR door and to provide indication when the door is not fully shut and latched.

VII. Previous Similar Events at the Same Site

LER 390/2017-014 reported an event where an individual left a MCR boundary door open. This was identified as a human performance error.

LER 390/2017-007-001 reported multiple instances over a three year period where a control room boundary door had been left open due to personnel error and promptly closed by operations in response to a low control room positive pressure alarm. The causes of these events are similar.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Watts Bar Nuclear Plant, Unit 1	05000390	YEAR	SEQUENTIAL NUMBER	REV NO.
		2018	- 003	- 00

NARRATIVE

VIII. Additional Information

None.

IX. Commitments

None.